

Problem 3.1*

Let (Ω, μ) be an outer measure space, suppose $u_n, u \in L^1(\mu)$ and $u_n \rightarrow u$ a.e. Show the following assertions are equivalent:

- (a) $\lim_{n \rightarrow \infty} \int_{\Omega} |u_n(x) - u(x)| \mu(dx) = 0.$
- (b) $\lim_{n \rightarrow \infty} \int_{\Omega} |u_n(x)| \mu(dx) = \int_{\Omega} |u(x)| \mu(dx).$

Solution.

Problem 3.2*

Let $f : [0, 1] \rightarrow [0, +\infty)$ be a $\mathcal{L}^1$ -measurable function such that $\int_0^1 f(x) dx < +\infty$. Suppose that for all $n = 1, 2, \dots$, $\int_0^1 f^n d\mathcal{L}^1 = \int_0^1 f(x) d\mathcal{L}^1$. Show that f must be equal almost everywhere to the characteristic function of a set $E \subseteq [0, 1]$.

Solution.

Problem 3.3*

Suppose μ is an outer finite measure on X and $f_1, f_2, \dots$ is a sequence of μ -measurable functions from X to $[0, +\infty)$ such that $\lim_{k \rightarrow +\infty} f_k(x) = +\infty$ for each $x \in X$. Prove that for every $\varepsilon > 0$ there exists a μ -measurable set E such that $\mu(X \setminus E) < \varepsilon$ and $f_1, f_2, \dots$ converges uniformly to $+\infty$ on E (meaning for every $t > 0$ there exists $n \in \mathbb{N}$ such that $f_k(x) > t$ for all $k \geq n$ and all $x \in E$).

Solution.

Problem 3.4*

Suppose $X \subseteq \mathbb{R}^d$ is a compact set and $(f_n)_n \subseteq C(X)$ is a sequence of functions such that $f_n \leq f_{n+1}$ for all natural numbers n . Assume

$$f(x) := \lim_{n \rightarrow +\infty} f_n(x) < +\infty \quad \forall x \in X.$$

Show that f is continuous if and only if $(f_n)_n$ converges uniformly to f .

Solution.

Problem 3.5*

Let μ and ν be two outer measures on Ω . We say that ν is absolutely continuous with respect to μ and we write $\nu \ll \mu$ if $\nu(E) = 0$ whenever $E \subseteq \Omega$ is ν -measurable and $\mu(E) = 0$. Assume $\nu[\Omega] < \infty$. Show that the following are equivalent:

- (a) $\nu \ll \mu$
- (b) For every $\varepsilon > 0$ there exists $\delta > 0$ such that $\nu(E) < \varepsilon$ whenever E is ν -measurable and $\mu(E) < \delta$.

Solution.

Problem 3.6*

A collection of functions $\{f_\alpha\}_{\alpha \in A} \subseteq L^1(\mu)$ is called uniformly integrable if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\int_E |f_\alpha(x)| \mu(dx) < \varepsilon$ for all $\alpha \in A$ whenever E is measurable and $\mu(E) < \delta$. Show that

- (a) any finite subset of $L^1(\mu)$ is uniformly integrable.
- (b) if $\{f_n\}_{n \in \mathbb{N}} \subseteq L^1(\mu)$ is a sequence that converges to f in $L^1(\mu)$ then $\{f_n\}_{n \in \mathbb{N}}$ is uniformly integrable.

Solution.

Problem 3.7*

Let $I = [-\frac{1}{2}, \frac{1}{2}]$. For each $x, y \in I$ we write that $x \sim y$ if $x - y \in \mathbb{Q}$. For $x \in I$ set $K(x) = \{y \in I : y \sim x\}$. Let $\{r_k\}_{k=0}^\infty$ be an enumeration of $[-1, 1] \cap \mathbb{Q}$ such that $r_0 = 0$.

- (a) Show that $\sim$ is an equivalence relation.
- (b) Let A be a set containing exactly one member of each equivalence class (the axiom of choice ensures existence of such an A) and set $A_k = A + r_k$. Show that

$$I \subseteq \bigcup_{k=0}^\infty A_k \subseteq \left[-\frac{3}{2}, \frac{3}{2}\right].$$

- (c) Conclude that A is not $\mathcal{L}^1$ -measurable if $\mathcal{L}^1$ is the one-dimensional Lebesgue measure. (Hint: compare $\mathcal{L}^1[A_k]$ to $\mathcal{L}^1[A]$.)

Solution.

Problem 3.8

Suppose μ is an outer measure on X . Prove that the following are equivalent:

- (a) The measure μ is σ -finite.
- (b) There exists an increasing sequence $X_1 \subseteq X_2 \subseteq \dots$ of μ -measurable sets such that $X = \bigcup_{k=1}^\infty X_k$ and $\mu(X_k) < \infty$ for each $k \in \mathbb{N}$.
- (c) There exists a disjoint sequence $Z_1, Z_2, \dots$ of μ -measurable sets such that $X = \bigcup_{k=1}^\infty Z_k$ and $\mu(Z_k) < \infty$ for each $k \in \mathbb{N}$.

Solution.

Problem 3.9

Let μ be an outer measure on a set X and let ν be an outer measure on a set Y . Define

$$\mathcal{P}_o := \{A \times B : A \text{ is } \mu\text{-measurable, } B \text{ is } \nu\text{-measurable}\}.$$

(a) Prove that if $A_1, A_2 \subseteq X$ and $B_1, B_2 \subseteq Y$ then

$$(A_1 \times B_1) \cap (A_2 \times B_2) = (A_1 \cap A_2) \times (B_1 \cap B_2)$$

and

$$(A_1 \times B_1) \setminus (A_2 \times B_2) = [(A_1 \setminus A_2) \times B_1] \cup [A_1 \cap A_2 \times (B_1 \setminus B_2)].$$

(b) Deduce that any finite union of elements of $\mathcal{P}_o$ can be written as a finite union of disjoint elements of $\mathcal{P}_o$.

(c) Deduce that any countable union of elements of $\mathcal{P}_o$ can be written as a countable union of disjoint elements of $\mathcal{P}_o$.

Solution.

Problem 3.10

Let μ be an outer measure on a set X and let ν be an outer measure on a set Y . Define $\mathcal{P}_o$ as in Exercise 3.9,

$$\mathcal{P}_1 := \{\cup_{i=1}^{+\infty} S_i : S_i \in \mathcal{P}_o\}, \quad \text{and} \quad \mathcal{P}_2 := \{\cap_{i=1}^{+\infty} S_i : S_i \in \mathcal{P}_1\}.$$

Let $\mathcal{F}$ be the collection of all sets $S \subseteq X \times Y$ such that the function $x \rightarrow \chi_S(x, y)$ is μ -measurable for ν -almost every $y \in Y$, and the function $y \rightarrow \int_X \chi(x, y) d\mu(x)$ is ν -measurable. Define

$$\rho[S] := \int_Y \left(\int_X \chi(x, y) d\mu(x) \right) d\nu(y), \quad (S \in \mathcal{F}).$$

Note in fact that if $S \subseteq X \times Y$ then there exists $R \in \mathcal{P}_2$ such that $S \subseteq R$ and $\mu \times \nu[S] = \rho[R]$.

(a) Use that $\mathcal{P}_o \subseteq \mathcal{F}$ to deduce that $\mathcal{P}_1 \subseteq \mathcal{F}$.

(b) Prove that if $T \subseteq X \times Y$ and $\mu \times \nu[T] = 0$ then $T \in \mathcal{F}$ and $\rho[T] = 0$.

(c) Prove that if $S \subseteq X \times Y$ is $\mu \times \nu$ -measurable and $\mu \times \nu[S] < +\infty$ then there exists $R \in \mathcal{P}_2$ such that $\rho[R \setminus S] = 0$ and $R \setminus S \in \mathcal{F}$.

Solution.